

Topics

- Euclidean Space
- Neighbourhoods and Open sets
- Limit points, Closed sets, Dense sets

Def A metric space is a set X with a function $d: X \times X \rightarrow \mathbb{R}_{\geq 0}$

s.t for all $p, q \in X$,

1) $d(p, q) = 0$ iff $p = q$ *non-degeneracy*

2) $d(p, q) = d(q, p)$ *symmetry*

3) $d(p, q) \leq d(p, r) + d(r, q)$ for all $r \in X$. *triangle inequality*

d is called a metric on X . Elements of X are called points.

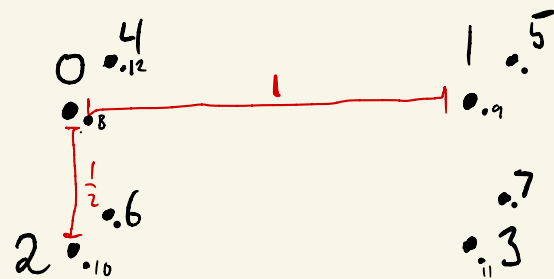
Example: $\mathbb{R}$ with $d(x, y) = |x - y|$.
"standard metric"

Example: $\mathbb{R}$ with $d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$
"discrete metric"

Example: 2-adic metric on $\mathbb{Z}$.

$d(a, b) = 2^{-n}$ where 2^n the

highest power of 2 dividing $a - b$.



Euclidean Space

$\mathbb{R}^d = \overbrace{\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}}^d$ is a d -dim vector space over $\mathbb{R}$.

Def Let $x = (x_1, \dots, x_d)$ $y = (y_1, \dots, y_d)$ be in $\mathbb{R}^d$.

- The dot product/inner product of x and y is

$$x \cdot y = \sum_{i=1}^d x_i y_i$$

- The norm of x is

$$|x| = \sqrt{x \cdot x} = \sqrt{\sum x_i^2}$$

Def Euclidean Space of dimension d is the metric space $\mathbb{R}^d$ with $d(x, y) = |x - y|$.

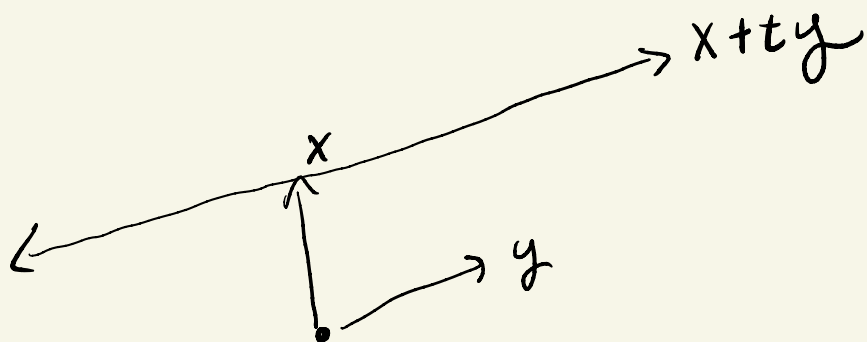
Theorem (R 1.37) For all $x, y, z \in \mathbb{R}^d$ and $\alpha \in \mathbb{R}$

- $|x| \geq 0$
- $|x| = 0$ iff $x = 0$
- $|\alpha x| = |\alpha| |x|$
- $|x \cdot y| \leq |x| |y|$
- $|x + y| \leq |x| + |y|$

Side note: A function $\mathbb{R}^d \rightarrow \mathbb{R}$ satisfying a, b, c, e is called a norm.

Proof (a), (b), (c) are easy.

(d) is the Cauchy-Schwarz inequality.



$$0 \leq (x+ty) \cdot (x+ty) = x \cdot x + 2x \cdot y t + y \cdot y t^2$$

$$\text{Recall: } ax^2 + bx + c \geq 0 \quad \text{for all } x \Rightarrow b^2 - 4ac \leq 0$$

$$(2x \cdot y)^2 - 4(x \cdot x)^2 (y \cdot y)^2 \leq 0$$

$$(x \cdot y)^2 \leq (x \cdot x)(y \cdot y)$$

$$|x \cdot y| \leq |x||y|$$

$$(e) \quad |x + y|^2 = (x + y) \cdot (x + y)$$

$$= x \cdot x + 2x \cdot y + y \cdot y$$

$$\leq |x|^2 + 2|x||y| + |y|^2$$

$$= (|x| + |y|)^2$$

Take square root:

$$|x + y| \leq |x| + |y|$$

Theorem $d(x, y) = |x - y|$ is in fact a metric.

Non-degeneracy:

$$|x - y| = 0 \text{ implies } x - y = 0 \text{ by 1.37b}$$

Symmetry:

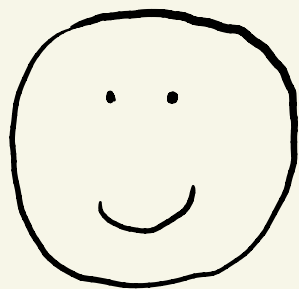
$$|x - y| = |(-1)(y - x)| = |-1| |y - x| = |y - x| \text{ using 1.37c}$$

Triangle inequality:

$$|x - z| = |(x - y) + (y - z)| \leq |x - y| + |y - z| \text{ using 1.37d}$$

Remark: any subset of a metric space is a metric space

$\mathbb{R}^2$



I'm a metric space!

Def Let $a < b$ be real numbers.

• $(a, b) := \{x \in \mathbb{R} \mid a < x < b\}$ is an open interval.

• $[a, b] := \{x \in \mathbb{R} \mid a \leq x \leq b\}$ is a closed interval.

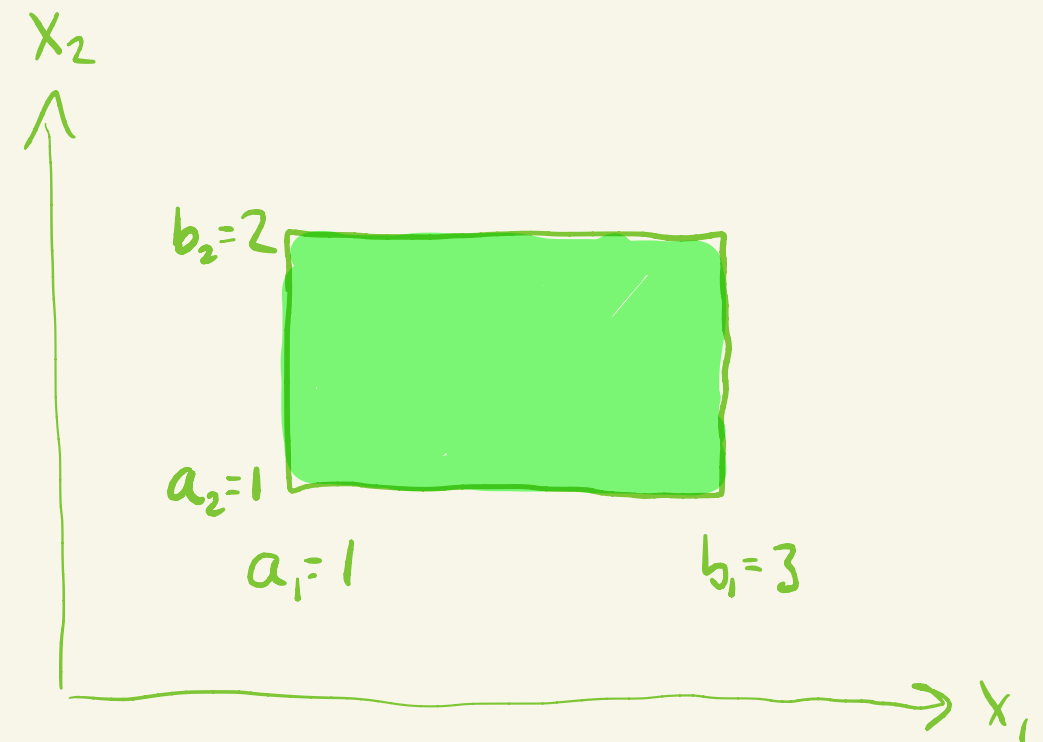
• $[a, b) := \{x \in \mathbb{R} \mid a \leq x < b\}$
 $(a, b] := \{x \in \mathbb{R} \mid a < x \leq b\}$ are half-open intervals.

Def Let $a_i < b_i$ for $i = 1, \dots, d$

$\{x \in \mathbb{R}^d \mid a_i \leq x_i \leq b_i\}$ is a d-cell.

"

$$[a_1, b_1] \times \dots \times [a_d, b_d]$$

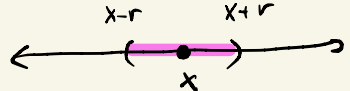


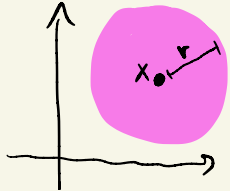
Def Let X be a metric space. Let $x \in X$ and $r > 0$.

$$N_r(x) = \{y \in X \mid d(x,y) < r\}$$

is the neighbourhood of x of radius r .

Examples

• $x \in \mathbb{R}$  $N_r(x) = (x-r, x+r)$

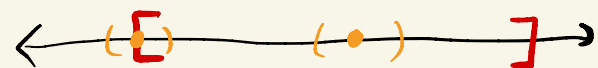
• $x \in \mathbb{R}^k$  $N_r(x) = \{y \in \mathbb{R}^k \mid \|x-y\| < r\}$
is called an open ball.

• for $X = (\mathbb{Z}, d_{2\text{-adic}})$ $N_{\frac{3}{4}}(0) = \{\text{even integers}\}$

Def $p \in E \subset X$ is an interior point of E if

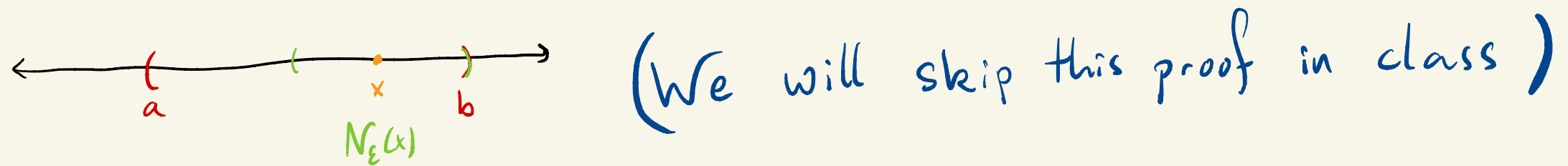
$\exists r > 0$ s.t. $N_r(p) \subset E$. The set of interior points is the interior, and is denoted E° .

Example: The interior of $[0, 1]$ is $(0, 1)$. The points 0, and 1 are not interior,



Def $E \subset X$ is open if every $x \in E$ is an interior point. That is, $E^\circ = E$.

Example: Every open interval (a, b) is open. Every point is interior. $(a, b)^{\circ} = (a, b)$.



Proof: Let $x \in (a, b)$. Let $\varepsilon = \min\{x - a, b - x\}$. By definition of (a, b) , $a < x < b$, so $\varepsilon > 0$. Claim: $N_{\varepsilon}(x) \subset (a, b)$. Suppose $y \in N_{\varepsilon}(x)$, i.e. $|y - x| < \varepsilon$. This implies $y - x < \varepsilon$, so $y < \varepsilon + x < (b - x) + x = b$. Also, $x - y < \varepsilon$, so $y > x - \varepsilon > x - (x - a) = a$.

Theorem Neighbourhoods are open in any metric space.

Let $p \in X$, $r > 0$.

Let $q \in N_r(p)$.

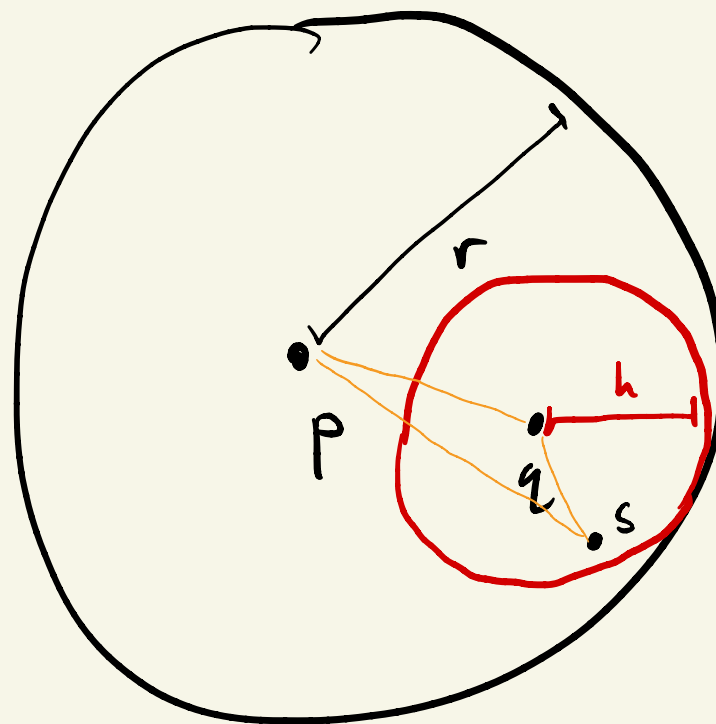
Let $h = r - d(p, q)$. Note that $h > 0$.

For all $s \in X$ s.t., $d(s, q) < h$, we have

$$d(p, s) \leq d(p, q) + d(q, s) < (r - h) + h = r$$

so $N_h(q) \subset N_r(p)$. So q is interior pt. of $N_r(p)$.

□



Def $p \in X$ is a limit point of a set $E \subset X$ if for all $r > 0$, $N_r(p)$ contains some $q \neq p$ s.t., $q \in E$. Let E' denote the set of limit points of E .

Example: $E = \{\frac{1}{n} \mid n \in \mathbb{N}\} \subset \mathbb{R}$. $0 \in \mathbb{R}$ is a limit point.

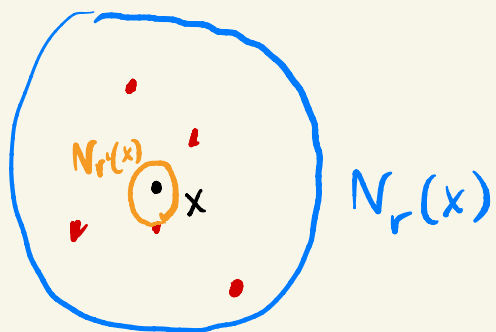
$$\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \dots \right)$$

For all $\varepsilon > 0$, $\exists n \in \mathbb{N}$ s.t. $n > \frac{1}{\varepsilon}$.

It follows that $\frac{1}{n} < \varepsilon$, so $\frac{1}{n} \in N_\varepsilon(0)$.

Theorem If x is a limit point of E , then $N_r(x) \cap E$ is infinite for all $r > 0$.

proof: suppose $N_r(x) \cap E$ is finite, but has points other than x .


$$\text{Let } r' = \min \{ d(x, y) \mid y \in N_r(x) \cap E \text{ s.t. } y \neq x \}.$$

$N_{r_1}(x) \cap E$ has no points not equal to x .

Corrolary • Finite subsets have no limit points.

Def $E \subset X$ is closed if every limit point of E is in E . That is, $E' \subset E$.

Examples:

1) Any finite subset of a metric space is closed.

2) $[a, b] \subset \mathbb{R}$ is closed.

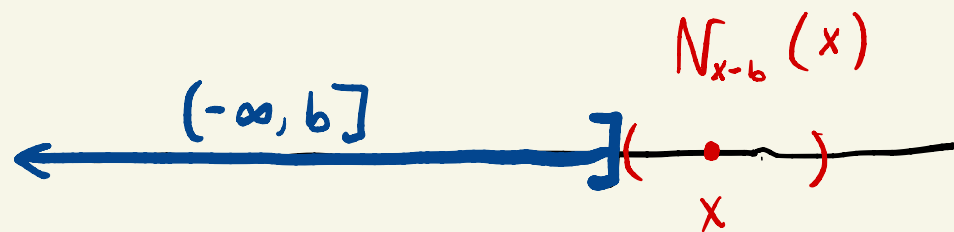
3) $(-\infty, b] = \{x \in \mathbb{R} \mid x \leq b\}$ is closed. So is $[a, \infty)$.

proof of 3: We show every $x \notin (-\infty, b]$ is not a limit point.

Suppose $x > b$. Let $y \in N_{x-b}(x)$.

$$y > x - (x - b) = b$$

So $y \notin (-\infty, b]$, so $N_{x-b}(x) \cap [a, b] = \emptyset$, so x is not a limit point.



Def $E \subset X$ is dense if every $x \in X$ is either in E or a limit point of E . That is, $E \cup E' = X$.

Ex

① X is dense in X for any metric space X .

② $(0,1)$ is dense in $[0,1]$.

③ $\mathbb{Q}$ is dense in $\mathbb{R}$.

proof of 3: Let $x \in \mathbb{R}$, and let $\varepsilon > 0$. There exists

some $q \in \mathbb{Q}$ s.t., $x < q < x + \varepsilon$ by our old "density of $\mathbb{Q} \subset \mathbb{R}$ " theorem.

This implies $q \in N_\varepsilon(x)$ and $q \neq x$. Conclusion: x is limit point of $\mathbb{Q}$.